

No. of Functional Features Included in the Solution

Date	1 July 2026
Team ID	SWTID-2026-8096
Project Name	Rising Waters
Maximum Marks	5 Marks

Functional Features Overview

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Rainfall Data Input	Allows users to enter annual and seasonal rainfall values for prediction.	User Interface	Done	High
2	Flood Prediction Model	Uses a trained Random Forest Machine Learning model to predict flood occurrence.	ML Prediction Engine	Done	High
3	Prediction Result Display	Displays whether there is a Flood Chance or No Flood Chance based on model output.	Result Module	Done	High
4	Responsive Web Interface	Provides an easy-to-use interface with responsive design across devices.	Frontend	Done	Medium
5	Rain Animation Background	Adds animated rainfall effect with background image to improve user experience.	UI Enhancement	Done	Medium
6	Input Validation	Validates user-entered rainfall values before prediction.	Backend Logic	Done	Medium
7	Model Integration with Flask	Integrates the trained ML model into the Flask web application.	Backend	Done	High

8	Cloud Deployment	Deploys the application online using Render for public access.	Deployment	Done	Medium
---	------------------	--	------------	------	--------

Feature Summary

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5
Additional / Nice-to-Have Features	3
Features Tested & Verified	8

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Rainfall Input Form, Responsive Design, Rain Animation
2	Backend / Logic	3	Flask Routing, Input Validation, Prediction Processing
3	Database / Storage	1	Trained Machine Learning Model (model.pkl)
4	API / Integration	1	Flask Integration with Random Forest Prediction Model
5	Security / Authentication	0	Not Implemented in Current Version

Notes

- The system predicts flood occurrence using rainfall parameters as input.
- A Random Forest machine learning model is integrated with the Flask backend.
- The application includes a responsive user interface with rainfall animation for improved usability.